

RealIntelligence - IQ



Context-Aware Analytics & Event Fusion Engine



RealIntelligence (IQ): Context-Aware Analytics & Event Fusion Engine

RealIntelligence (IQ), added to RealVue, turns video surveillance into a proactive intelligence platform without vendor lock-in. It fuses best-of-breed video engine and AI skillsets—RealTrack for advanced object tracking, RealFace for embedded facial recognition, and RealPlate for high-performance LPR/OCR—with external data from 167+ POS systems, access control, alarms, scales, GPS and more.

Deployable in minutes on existing hardware, IQ delivers real-time threat detection, forensic search, and automated alerts. With custom models and logical rules, it enables tailored workflows—transforming “dark data” into actionable metadata. Backed by Realwave’s 24/7 Pro Services, IQ cuts false alarms by 90%, reduces manual investigations by 80%, and enables sub-second responses—improving team effectiveness by 10X.

Key Features

- **AI Skillsets:** RealTrack for deep learning-based tracking of people, vehicles, hands, bags, or custom objects through occlusions; RealFace for NIST-ranked facial recognition (99.4% accuracy, even with masks) supporting age/gender grouping and remote databases; RealPlate for 99%+ accurate LPR/OCR on license plates, trailer/container IDs, DOT numbers, with make/model/color/pose classification at speeds up to 225 km/h.
- **Data Fusion & Automation:** Integrates IoT/POS for context-aware insights; Rules Engine triggers real-time alerts/actions for exceptions like refund fraud or unauthorized access, with logical rules for bespoke behavior detection.
- **Advanced Detection:** Intrusion, area entry/exit, loitering avoidance, object left behind/guarding, tailgating, line crossing, armed person (weapon) detection, fallen person alerts, crowd measurement/detection, and queue analysis for bottleneck identification.
- **Forensic & Appearance Search:** Quickly locate events using appearance-based filters (e.g., search by shirt/pants color, object type) or metadata across video/POS/IoT; supports keyword, bookmark, or smart motion queries for seconds-fast retrieval.
- **Plug-and-Play Integration:** Fully embedded in RealVue client for seamless configuration; no new hardware required—works with RGB/IR cameras, servers, storage; one-click false positive reporting for continuous model improvements.
- **Secure Real-Time Monitoring:** Analytics for low-light/complex environments; minimizes false alarms with accurate models; enables tailored reporting workflows via SMS/email/API.

Technical Excellence & Strategic Benefits

Technical Highlights

- **Analytics & AI Features:** Zones/lines, tamper detection, 3D calibration, appear/disappear filters, stopping/dwell/direction/speed filters, abandoned/removed objects, aggressive behavior, fall detection, hand-object, counting (people/vehicles, even in dense scenes).
- **Specialized Capabilities:** Weapon detection via armed person alerts; crowd measurement for density estimation; queue analysis through dwell metrics and loitering filters; appearance search for visual attributes like clothing color.
- **Event Rules & Interfaces:** Basic/advanced rules (tripwires, intrusion, tailgating); event sources (analytics, loss of signal, HTTP); actions (email, TCP, HTTP); video input (RTSP, files: avi/mp4/mpg); output (events via TCP/HTTP/SMTP, JSON metadata).
- **Hardware & Deployment:** Designed for diverse combos (Intel/AMD x86, ARMv8); stand-alone server mode streams from cameras/VMS; easy integration with VMS/NVR/PSIM via HTTP/TCP/email; HTML5 UI for configuration on desktop/mobile (Chrome 130+).

Benefits

RealIntelligence (IQ) eliminates siloed data and manual probes, delivering unified, exception-based insights with no vendor lock-in. Its open architecture integrates seamlessly with existing systems, reducing false positives and enabling real-time monitoring across enterprises. Tailored analytics workflows fuse video with business context for outcomes like 60% retail shrink reduction or 99% logistics accuracy, while forensic tools cut scrubbing time by 80%. Scalable from small sites to global hubs, IQ ensures privacy-first detection (e.g., anonymous LiDAR tracking) and compliance. Trusted for over 1.5 million channels, it drives ROI through predictive fraud detection and monetized data—backed by Realwave’s managed services for effortless deployment and 24/7 optimization.

Specifications

Feature	Details
AI Models	Object tracking, face recognition (NIST 1:1/1:N), LPR/OCR (99%+ accuracy)
Detection Types	Intrusion, loitering, tailgating, line crossing, armed/fallen person, crowd/queue analysis, demographics, appearance, weapons, fire, drones
Search Capabilities	Forensic/appearance-based (color attributes, metadata); seconds / petabytes
Platforms	Windows/Linux/macOS/Android; Intel x86/ARMv8; NVIDIA Jetson compatible
Integration	Access, PID, POS, IoT, VMS/NVR, REST API, SDKs, HTTP/TCP/email
Security	SOC2 Type 2, encrypted metadata; anomaly detection for threats